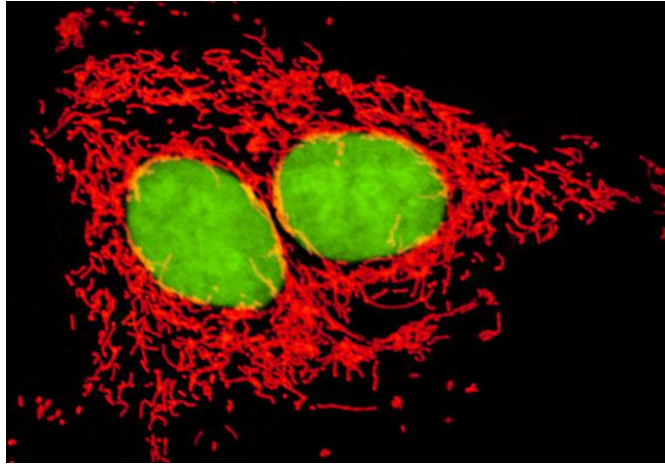


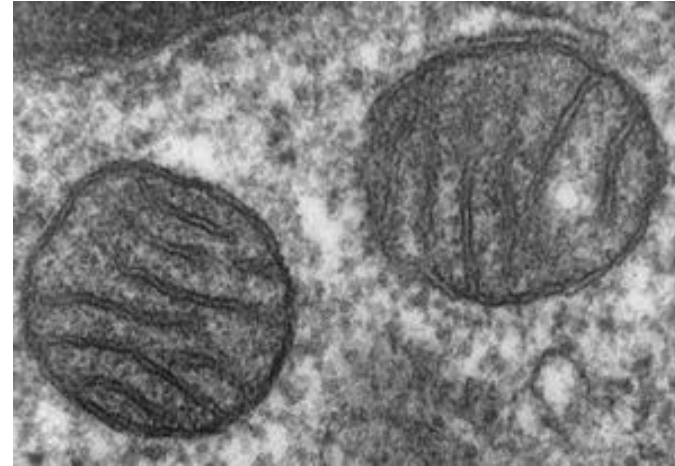
Mitochondrion

FLUORESCENCE MICROSCOPY (MITOTRACKER)

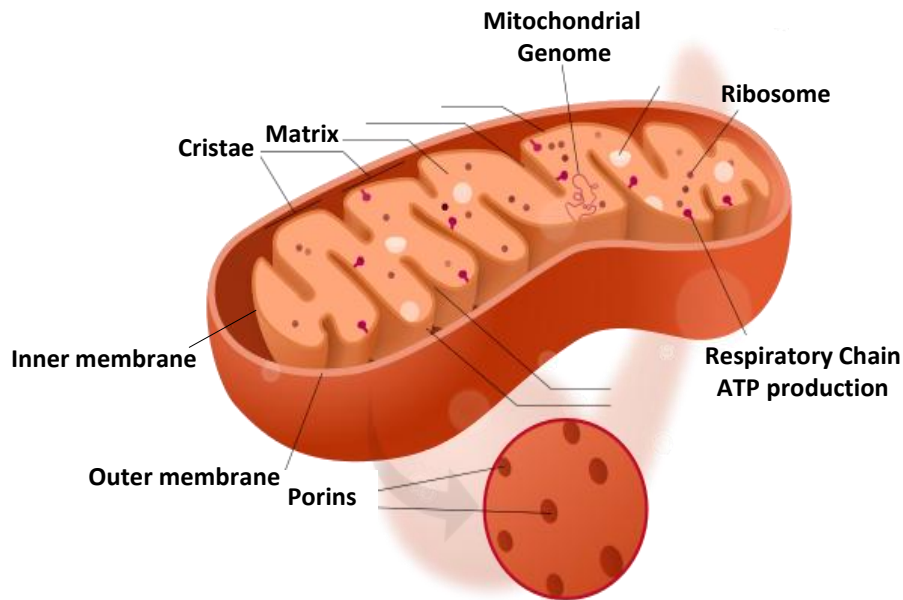


Number of mitochondria/cell is tissue dependent
Red blood cells 0/hepatocytes 2000

ELECTRON MICROSCOPY

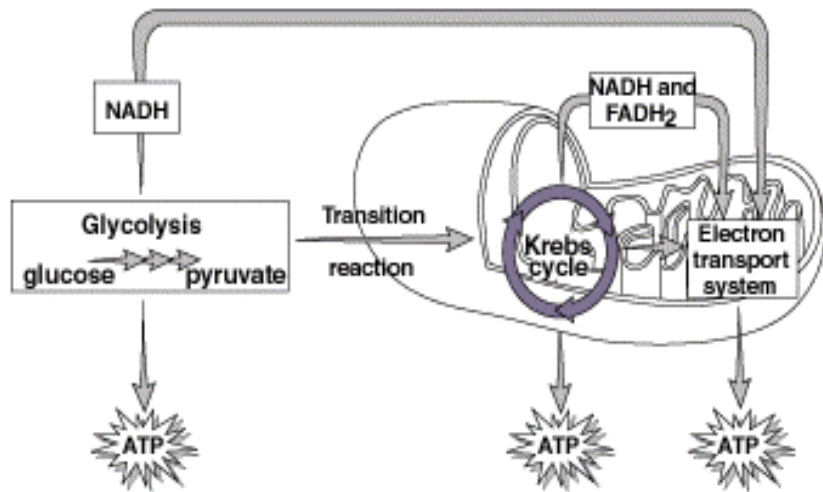


Size of mitochondria ranges between
0.5 and 2 μm in diameter



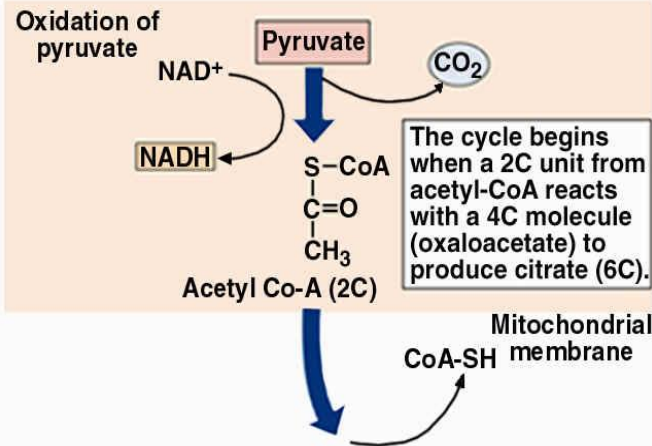
- Mitochondria "the powerhouse of the cell" are the primary source of ATP in the cell. Other important functions are intracellular Ca^{2+} homeostasis, cell cycle control, mitophagy and apoptosis.
- They are inherited only through the maternal oocyte.
- Distribution of mitochondria in the two daughter cells during mitosis is random.
- Their amount inside a cell is strictly dependent by the levels of energy consumption.
- They have their own ribosomal apparatus for protein synthesis.

Glycolysis & Krebs cycle: production of ATP+NADH/FADH₂



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Krebs Cycle Begins

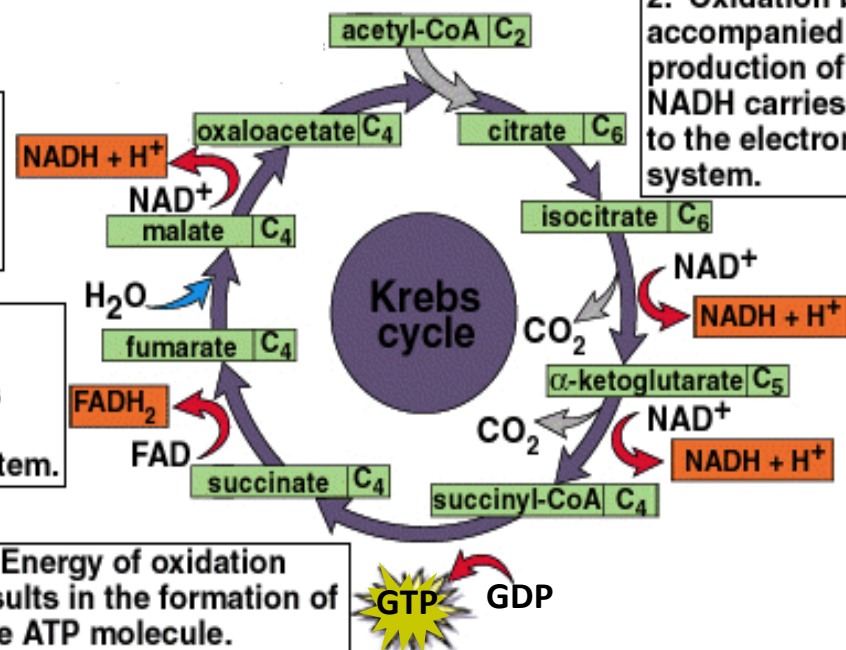


5. As the cycle returns to its starting point, another NADH is formed. NADH carries electrons to the electron transport system.

4. Oxidation by FAD produces FADH₂. This coenzyme also carries electrons to the electron transport system.

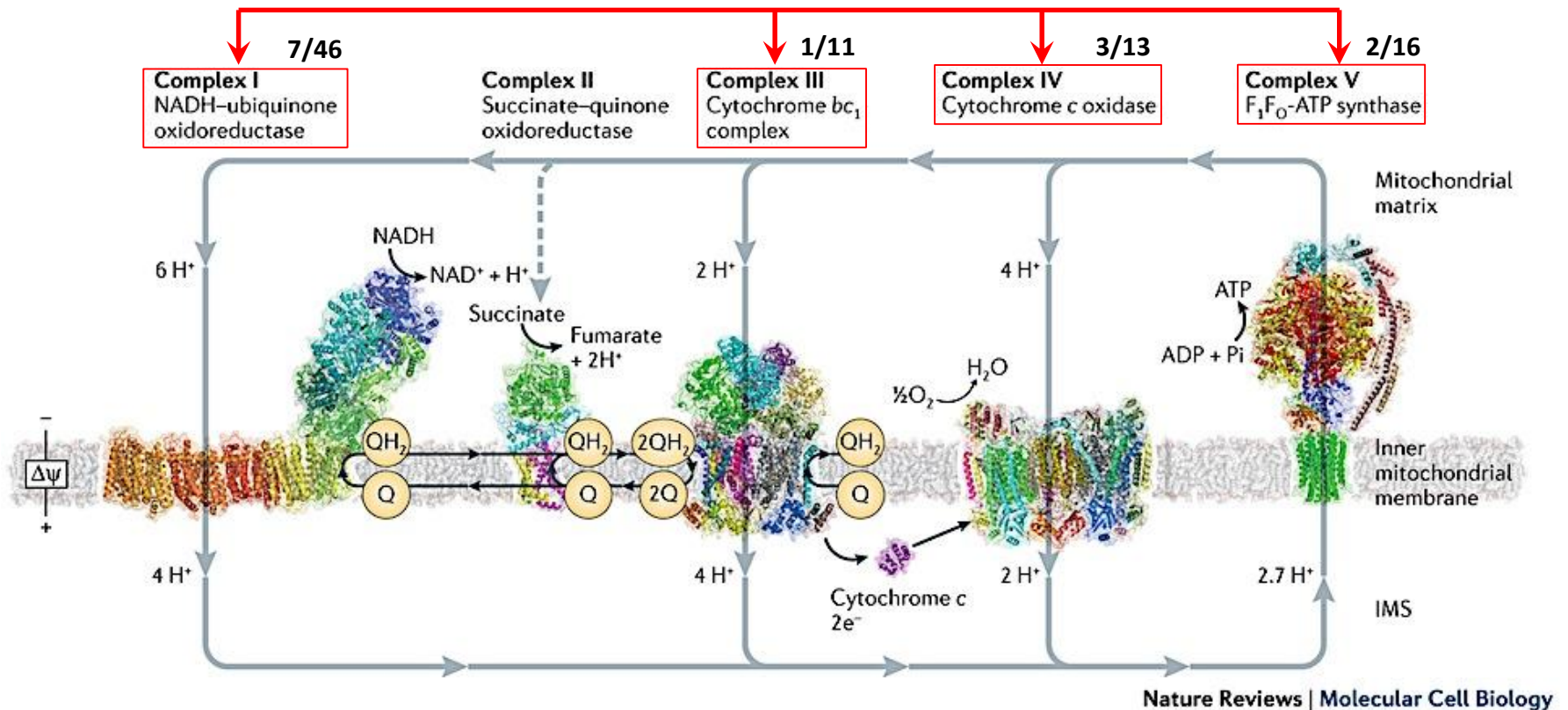
3. Energy of oxidation results in the formation of one ATP molecule.

2. Oxidation by NAD⁺ accompanied by the production of CO₂. NADH carries electrons to the electron transport system.



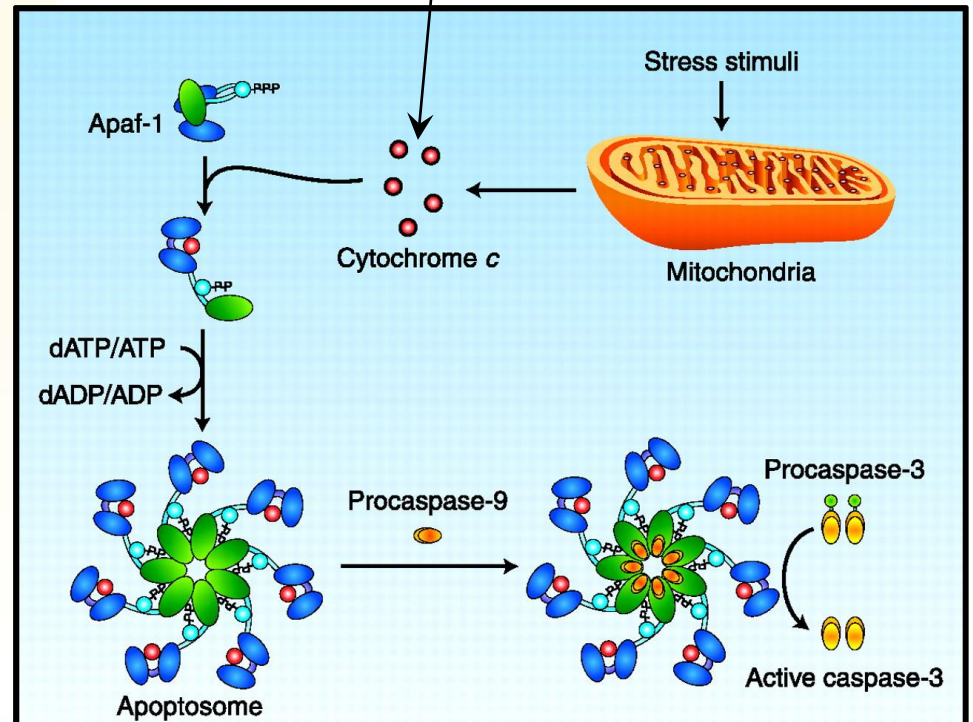
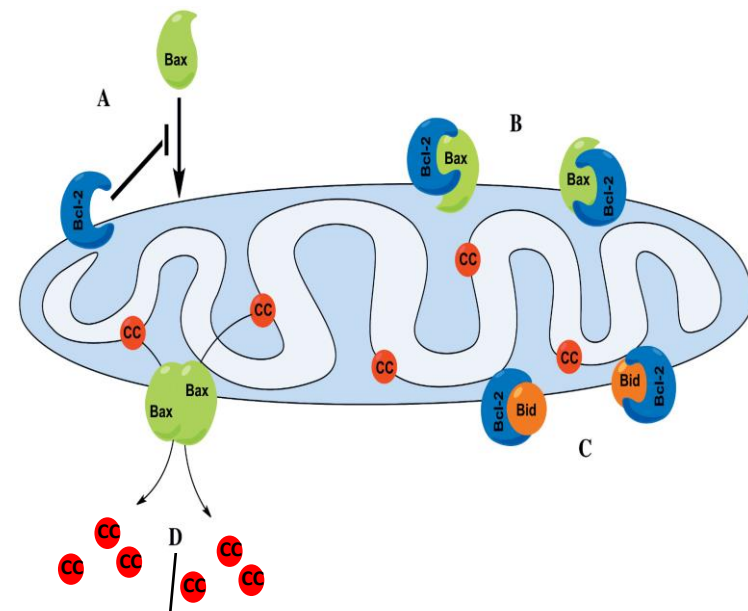
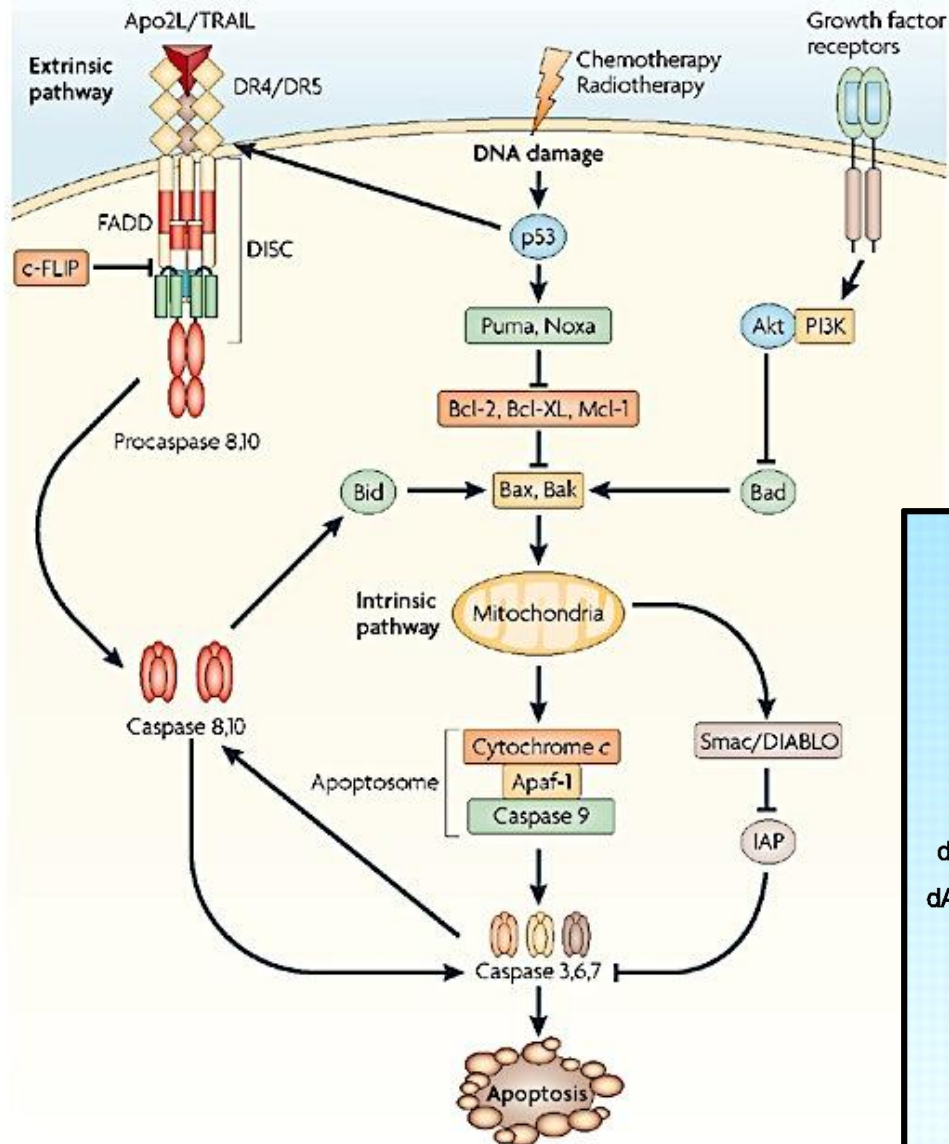
Respiratory chain and oxidative phosphorylation

Complex subunits encoded by mtDNA

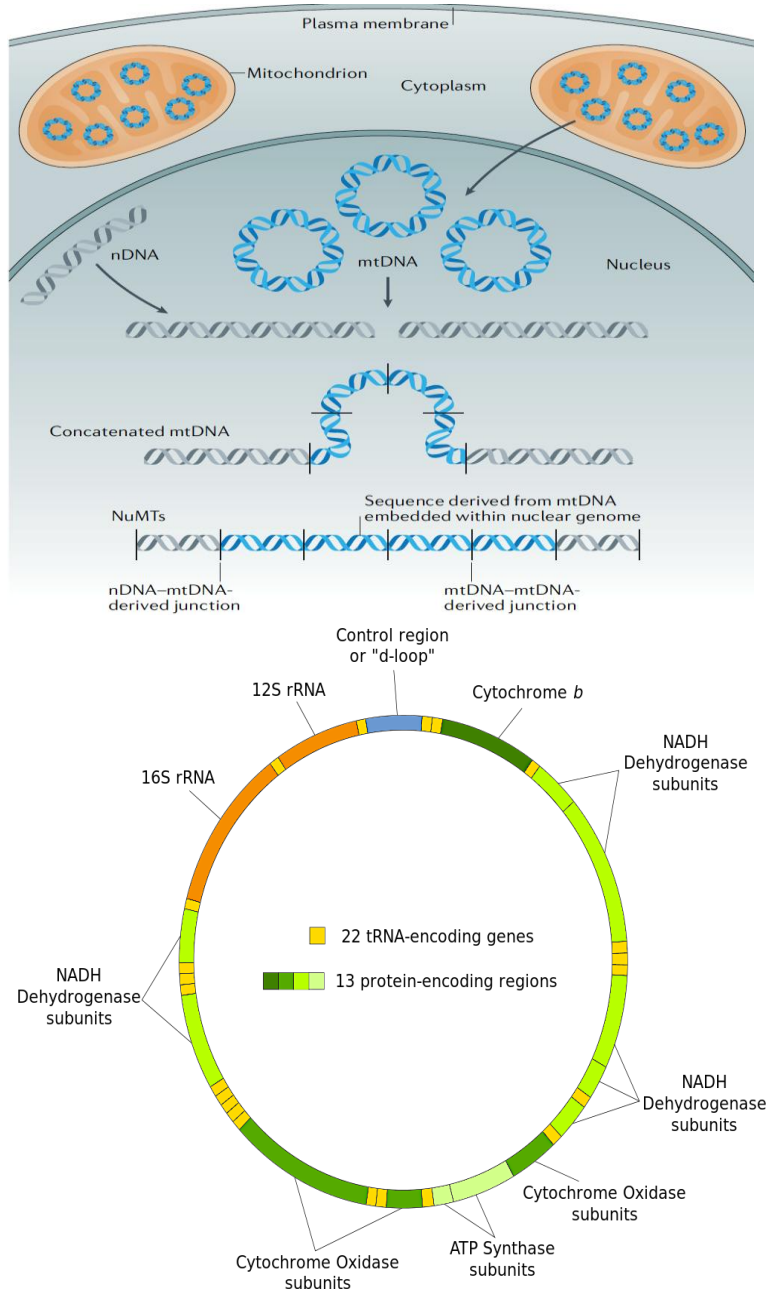


The mammalian mitochondrial electron transport chain (ETC) includes the proton-pumping enzymes complex I (NADH-ubiquinone oxidoreductase), complex III (cytochrome bc₁) and complex IV (cytochrome c oxidase), which generate proton motive force that in turn drives F₁F₀-ATP synthase. Electron transport between complexes is mediated by membrane-embedded ubiquinone (Q) and soluble cytochrome c. Complex I is the entry point for electrons from NADH, which are used to reduce Q to ubiquinol (QH₂). QH₂ is subsequently used by complex III to reduce cytochrome c in the intermembrane space (IMS), and complex IV uses cytochrome c to reduce molecular oxygen, which is the ultimate electron acceptor. **For each NADH molecule oxidized, 10 protons are translocated across the membrane** from the matrix to the IMS. Complex II (succinate-quinone oxidoreductase) provides an additional entry point for electrons into the chain.

Mitochondrial Apoptotic Pathway



~2.5 billion years, following the endosymbiosis of a proteobacterial ancestor



Mitochondrial Genome

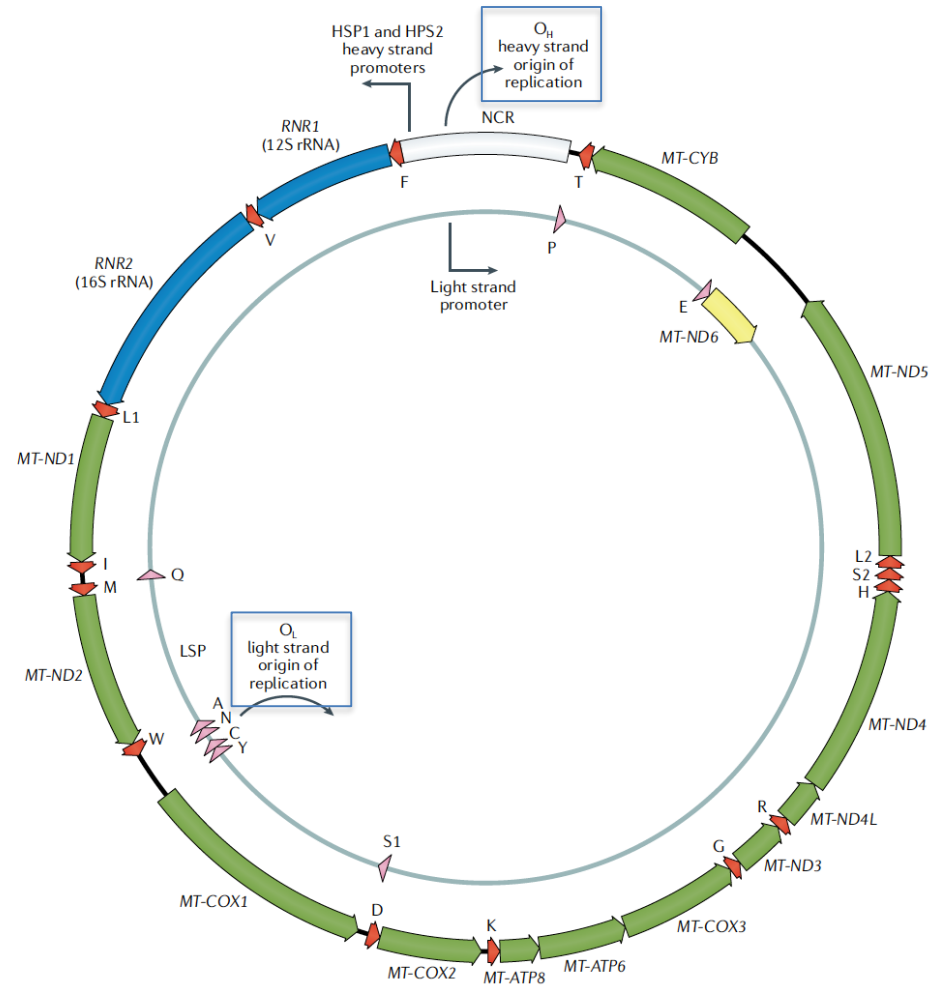
- Genes controlling several key functions have been passed to the host cell.
- **Circular and in multiple copies (from 2 to 10 x mito; 1.000-10.000 x cell).**
- Each molecule is approximately 16.5 Kb in length.
- Chromatin/Nucleosomes organization is totally absent.
- mtDNA replication is completely independent by nuclear DNA replication.
- The entire DNA molecule is bound by a nuclear encoded protein "Mitochondrial transcription factor A" (TFAM).
- TFAM-mtDNA interaction is essential to stabilize and protect mtDNA.
- mtDNA contains
 - 22 tRNA encoding genes
 - 16S and 12S ribosomal RNAs
 - 13 genes whose protein products are involved in respiratory chain and oxidative phosphorylation (out of 87).
 - a "D-Loop" region controlling both transcription and replication (~1100bp).
- mtDNA is transcribed as a unique polycistronic mRNA
- UV light, tobacco smoking, and ROS are major causes of mtDNA damage. mtDNA replication is source of age-related mutations.
- **BER, NER, MMR, and NHEJ pathways are imported from the nucleus into the mitochondrion for DNA damage repair.**

Mitochondrial Genome Replication

A human postmitotic neuron containing ~10,000 mtDNA molecules and living for 80 years will replace its entire mtDNA content ~3,000 times.

mtDNA replication begins with the leading strand emerging from the D-loop at the origin of heavy-strand replication (OH). Replication proceeds approximately two-thirds of the way around the molecule, exposing single-stranded DNA, before triggering synthesis of the lagging strand from the origin of light strand replication (OL).

The extended time spent in a single-stranded state has long been argued to be the source of the observed strand-specific mutation bias observed for mtDNA.



Mitochondrial Genome Alteration

Due to a lack of protective histones, high rate of replications, limited DNA repair activities, and proximity to ROS, mtDNA has a higher mutation rate than nuclear DNA (~10- to 200-times).

Point mutations

Approximately 60% of cancers harbors somatic point mutations of mtDNA. A large majority of such mutations are enriched in tumour compared to normal cells and hits non-conserved nucleotides.

25% of them, however, are a missense, no-sense, or frame-shift mutation of the coding mRNA, tRNAs, or the D-Loop.

Small genomic deletions

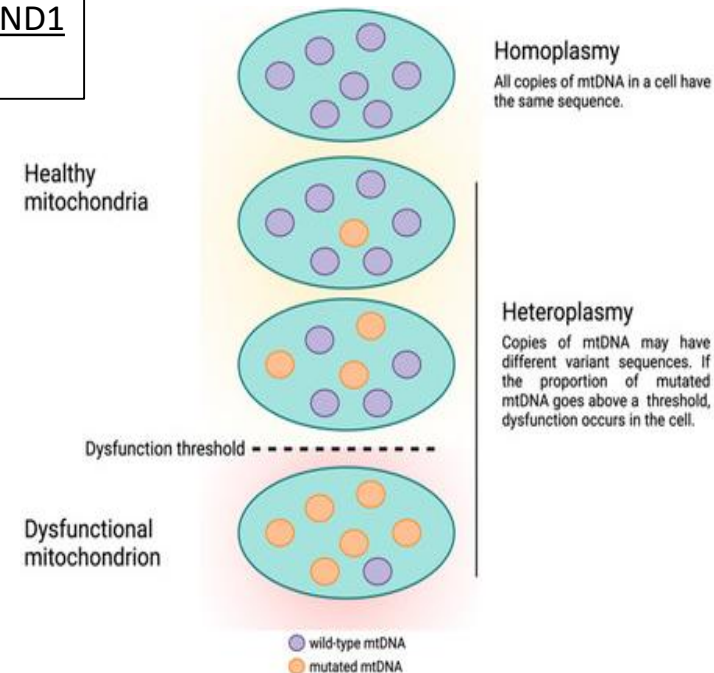
Small mtDNA deletion of tens of bases have been demonstrated to be robustly associated with different types of tumour. Among the most frequent, the 50-bp deletion flanked by a 9-bp direct repeat in the D-loop region, the 66-bp deletion flanked by the 8-bp direct repeat in the NADH Dehydrogenase 1 (ND1) gene, the 294-bp deletion in the ND1 gene.

Copy number variation

CNV has been demonstrated in several types of human cancer. Mechanisms are still unclear. This condition can depend on dysfunction of nuclear genes involved in mtDNA replication (mTAF, PolG, Twinkle, mtSSB), or by D-Loop mutations.

A mixed population of mitochondrial genomes: '**heteroplasmy**';
a homogeneous population of mtDNA: '**homoplasmy**'

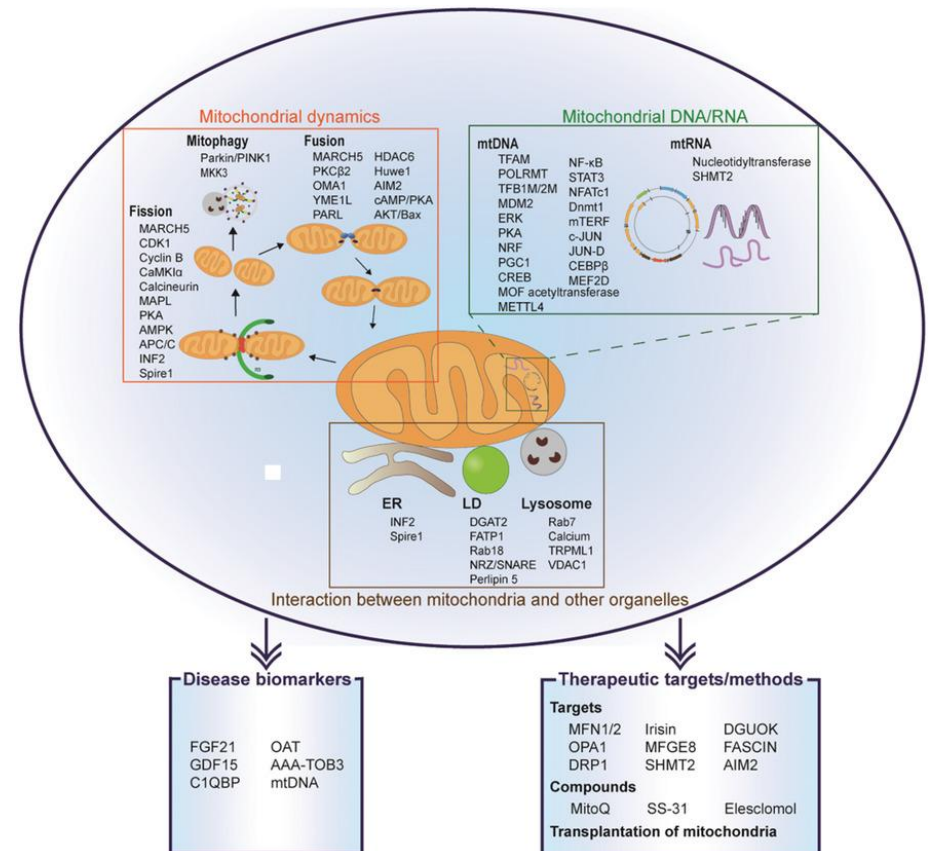
- some mtDNA mutations undergo preferential replication, either because they copy faster or because they become available to replicate more frequently.
- changes in allele frequency can occur when mtDNAs are partitioned during cell division, where the unequal distribution of different alleles leads to daughter cells with different heteroplasmic levels.



Mitochondrial genome copy number variation as potential circulating cancer biomarker

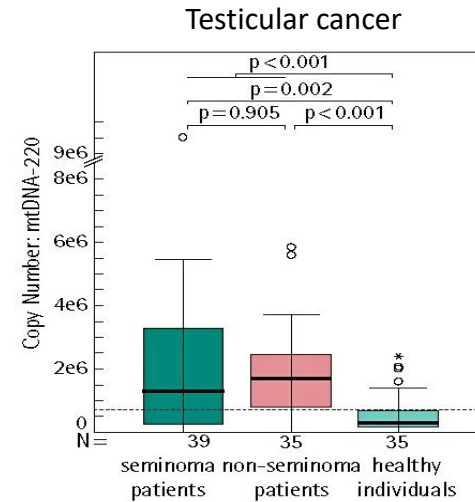
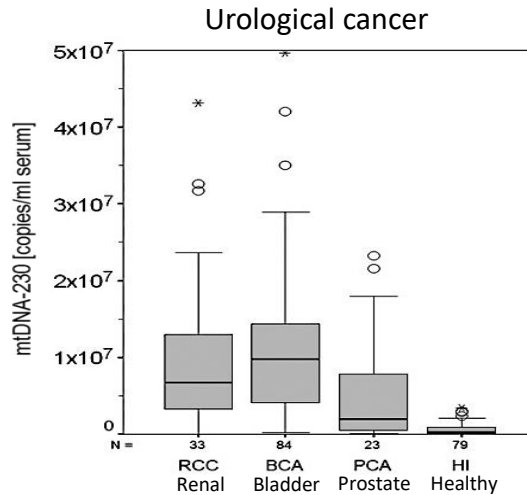
Monitoring aberrant mtDNA alterations is rapidly emerging as a compelling molecular tool for early tumor detection with many advantages over the nuclear DNA-based methods.

- The mitochondrial genome is short in length and possesses simple organization, which make the genome-wide screening of mtDNA much easier and more cost-effective than employing nDNA.
- The higher abundance of mtDNA molecules relative to nDNA can drastically enhance the sensitivity and accuracy of using mtDNA variations as a potential biosensor to distinguish rare malignant cells, even single cells, in circulation.
- Circulating cell-free mtDNA can be readily detectable in easily procured bodily fluids (e.g. peripheral whole blood, salivary rinse, urine, sperm, nipple aspirate fluid).



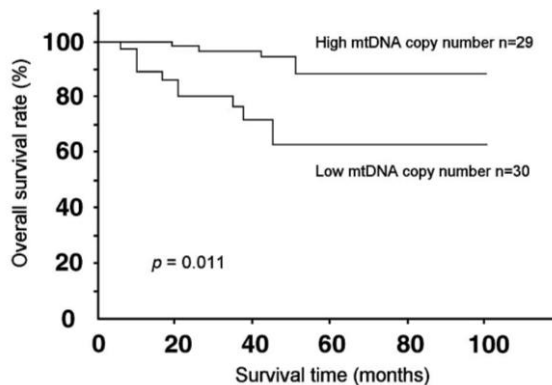
Mitochondrial Genome CNV

Diagnosis

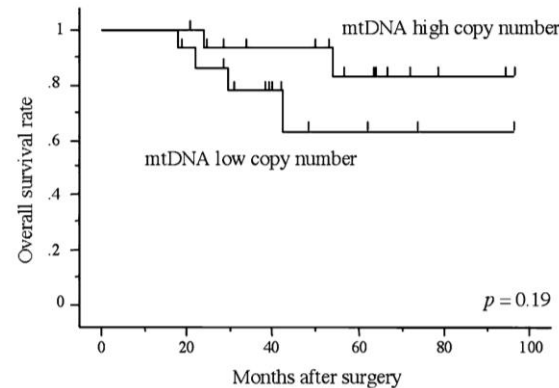


Prognosis

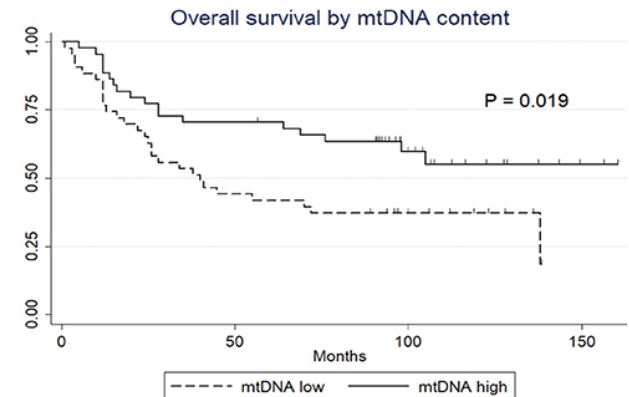
Breast Cancer



Hepatocellular Carcinoma



Meningioma



mtDNA low CN

High grade/undifferentiated/Mets
 Gastric, Ovarian, Prostate
 and Colon cancers

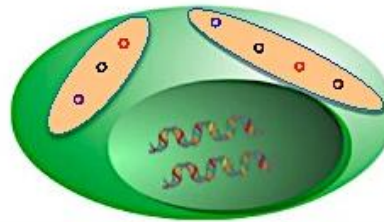
mtDNA high CN

High grade/undifferentiated
 Head and Neck carcinoma

Heteroplasmy, homoplasmy and fermentation

Ovarian, Breast, Prostate, Gastric, Hepatocellular, and Kidney cancers

Parental Cells of Origin (respiratory active)



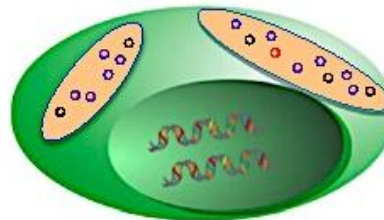
heteroplasmy

Two or more slightly different sequences of the mitochondrial genome coexist in a single cell.

Carcinogenesis:

with a transient amplification of the mitochondrial genome content

Cancerous Cells (respiratory active)



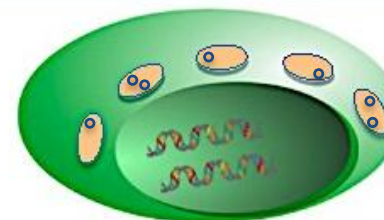
homoplasmy

Mutant mitochondrial genome largely overcomes wild type counterpart.

Cancer Progression:

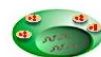
with a reduction-to-depletion of the mitochondrial genome content

Advanced Cancerous Cells (fermentative)



Depletion of the mitochondrial genome generates de(un)differentiated cells that harbors small and immature mitochondria localized around the nucleus with impaired MRC activity, similar to embryonic stem cells (ESCs).

(key)



Eukaryotic Cell



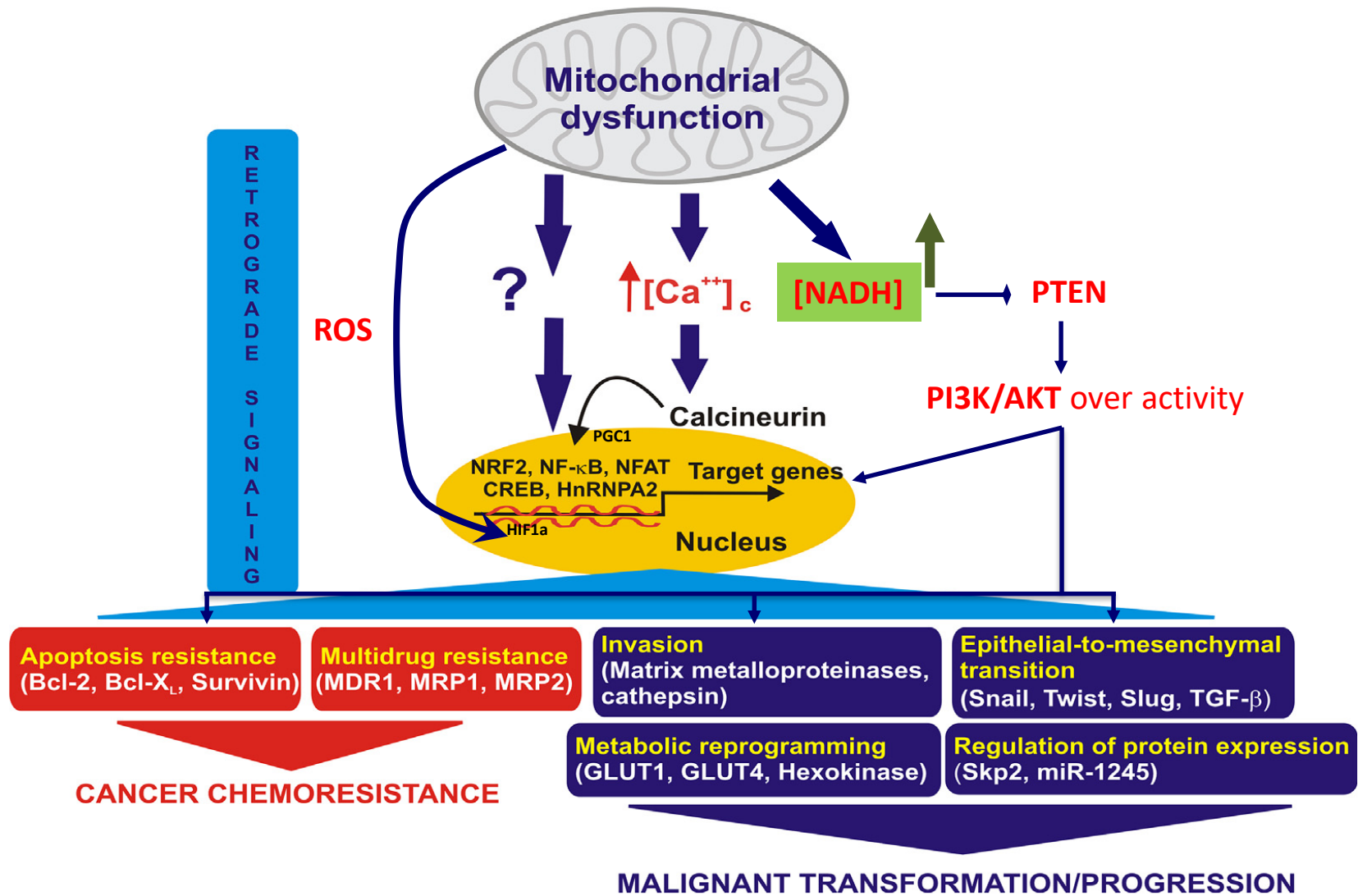
Mitochondrion

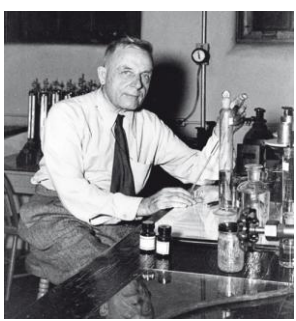


Mitochondrial Genome (Copy Number)

To this date, no study has elicited a mechanism that links the mitochondrial genome content to a mitochondrial generated signal promoting cancer progression.

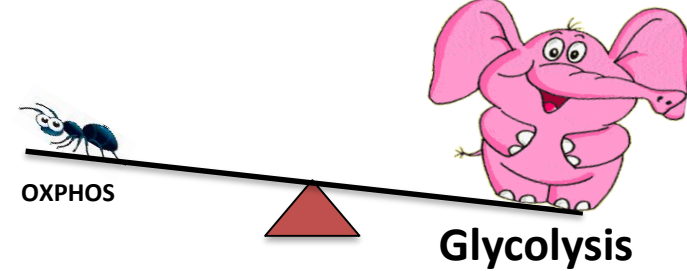
Mitochondrial genome alteration and response to therapy



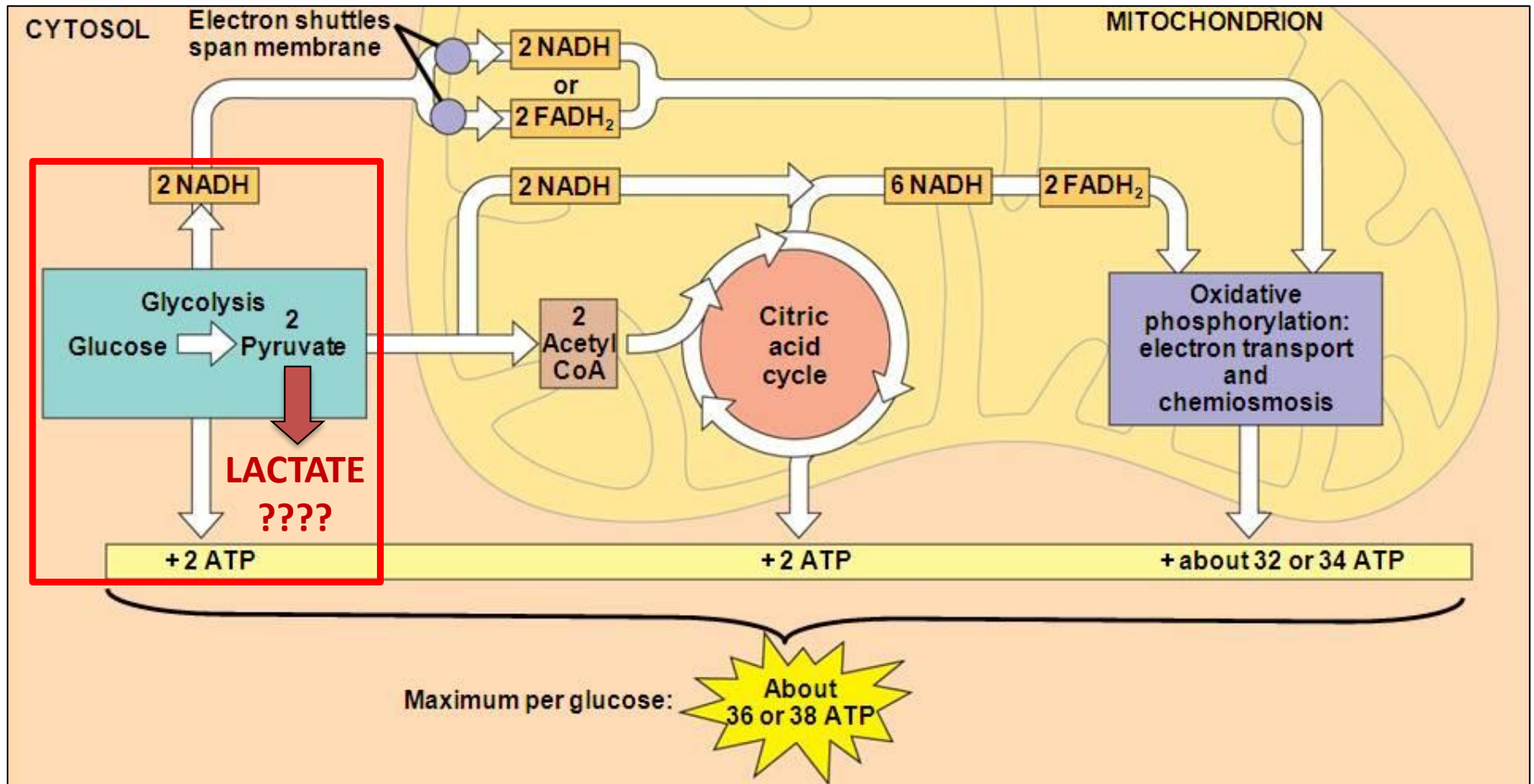


The Paradox of Cancer Metabolism

The Warburg Effect

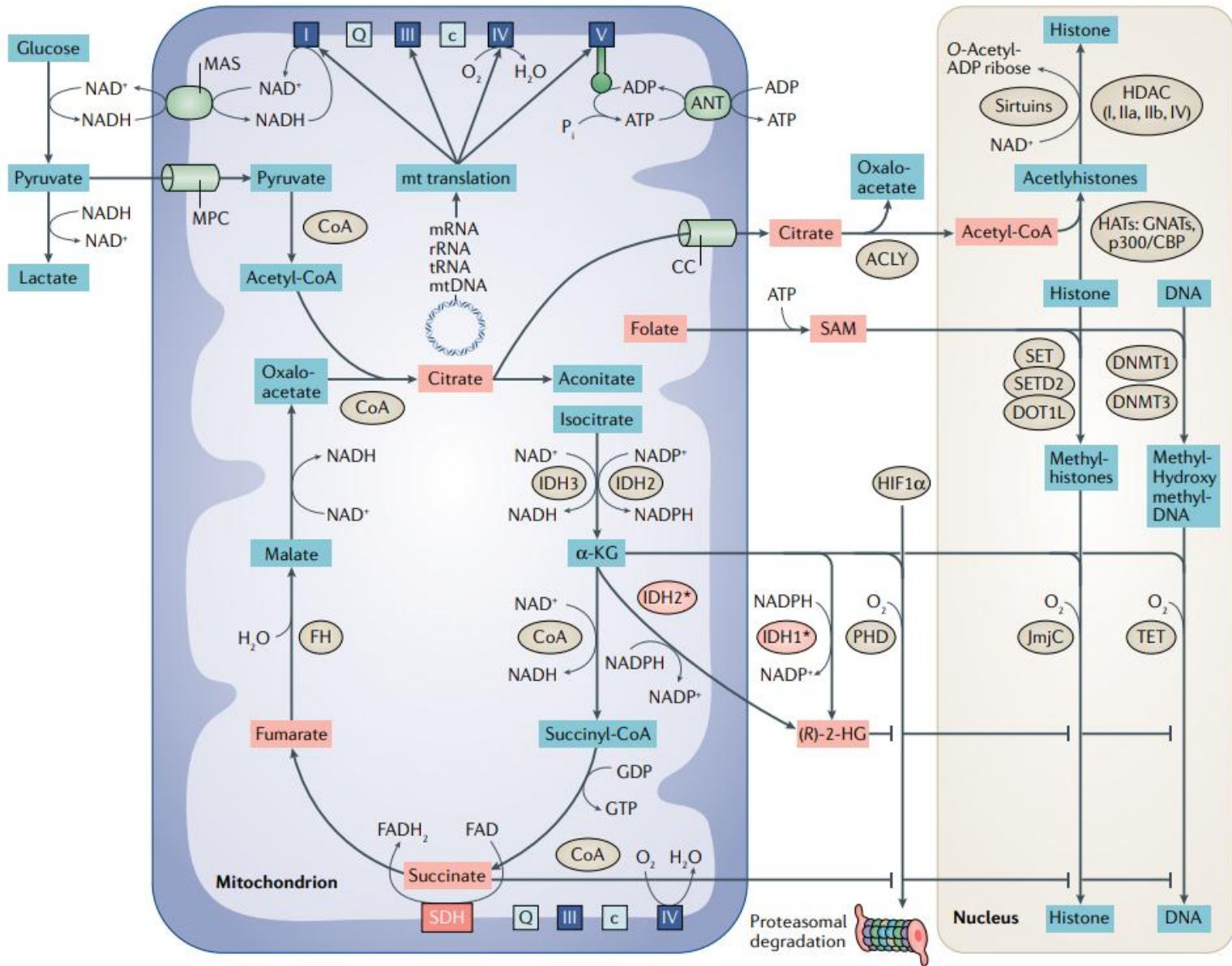


Cancer Cells



Normal Cells

Mutations in nDNA mitochondrial genes modulate the cancer epigenome



Article

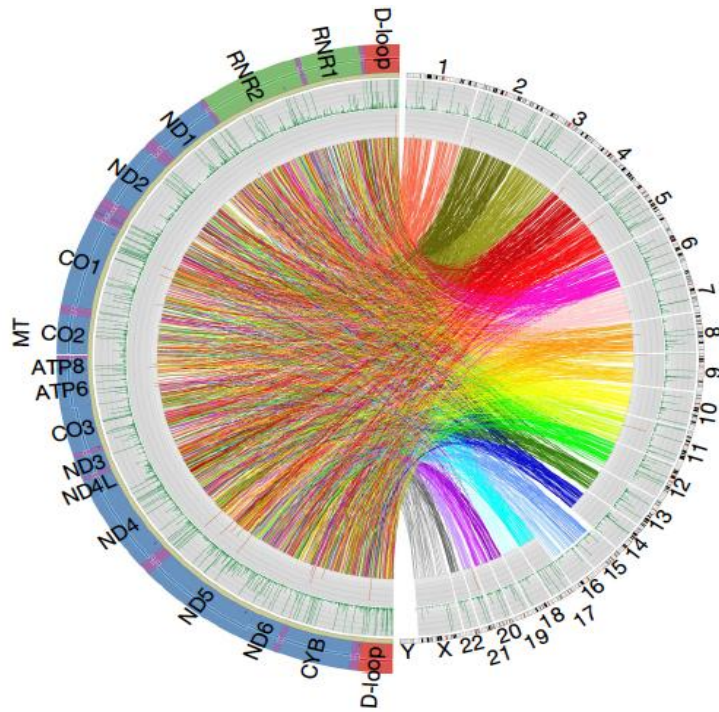
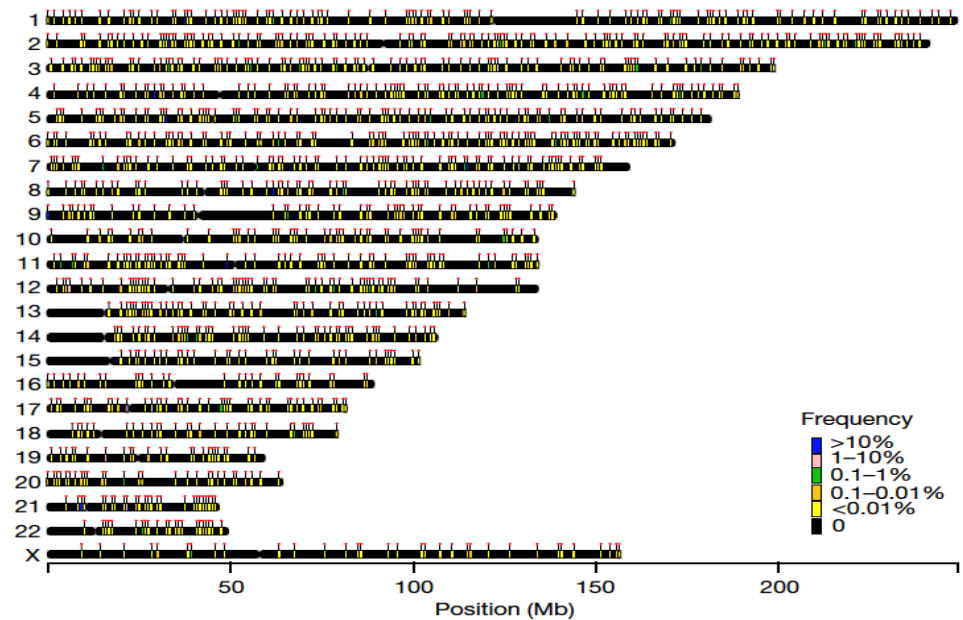
Nuclear-embedded mitochondrial DNA sequences in 66,083 human genomes

This process has left fragments of non-expressed mitochondrial DNA (mtDNA) throughout the non-coding space, with many NUMTs being shared across species, reflecting their ancient origin.

The majority of NUMTs were short insertions (63.2% of NUMTs were less than 200 bp and 77.8% were less than 500 bp in size). Major differences in the frequency and distribution of NUMTs between different ethnic groups were observed, with African and East Asian individuals being the most distinct in relation to the NUMT frequencies and chromosomal locations involved.

Whole genome sequences from 66,083 people—including 12,509 people with cancer

254,195 NUMTs that are not present in the reference sequence



Overall, we observed a strong positive correlation between the length of each chromosome and the number of NUMTs detected.

However, chromosomes 3, 6 and 21 had a larger number of NUMTs per Mb than the remaining autosomes. The Y chromosome was not analysed owing to the complex duplicated structure limiting confident alignment.

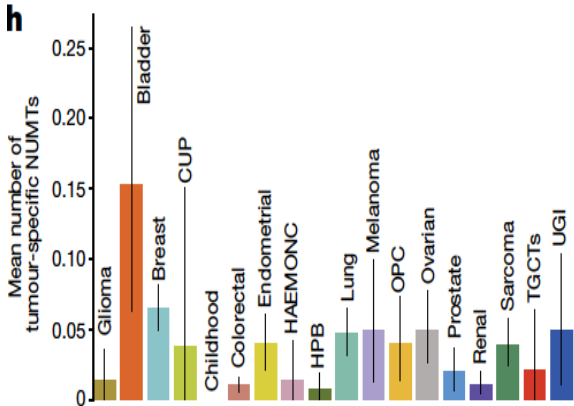
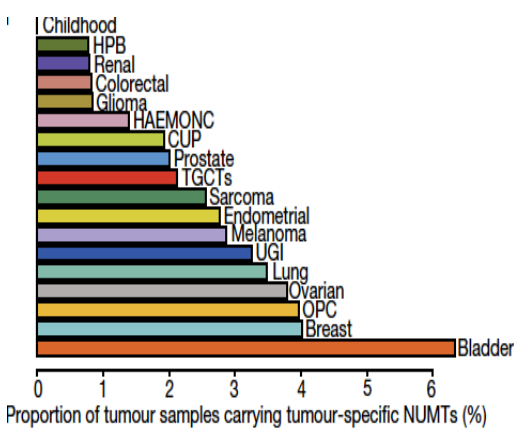
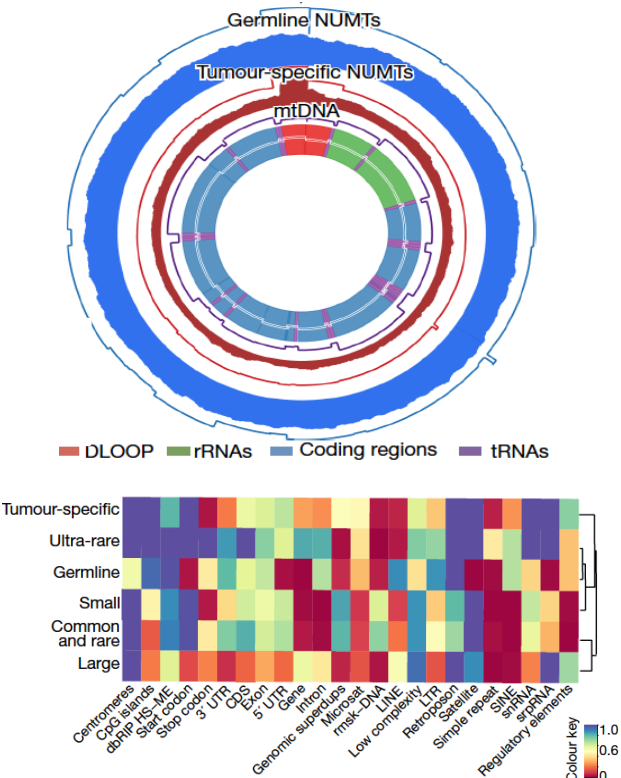
Common and rare NUMTs ($F \geq 0.1\%$) were more likely to occur near or within genomic duplications, while ultra-rare NUMTs were enriched in repetitive elements, simple repeats, SINES, and introns, consistent with selection against NUMTs that disrupt gene function.

Article

Nuclear-embedded mitochondrial DNA sequences in 66,083 human genomes

The signature of de novo NUMTs in cancer

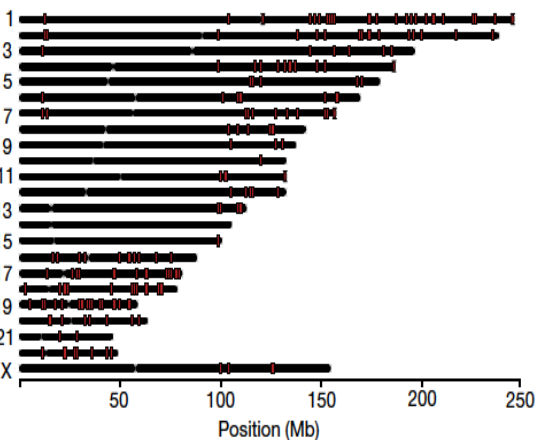
Overall, tumours have a higher mean number of NUMTs that were not present in the reference sequence of the corresponding normal tissue. The mtDNA segments forming de novo tumour NUMTs differed from those in the germline. They were less likely to involve MT-CO3 ($P = 7.7 \times 10^{-3}$), MT-ND4 ($P = 3.1 \times 10^{-3}$), MT-ND4L ($P = 3.4 \times 10^{-3}$) and MT-ND5 ($P = 5.3 \times 10^{-3}$), but more than 2.5-fold more likely to involve the D-loop ($P = 3.36 \times 10^{-36}$).



The proportion of tumours with a de novo NUMT depended on the tumour type, with renal and colorectal tumours having fourfold fewer NUMTs than breast cancers ($P = 1.93 \times 10^{-6}$) and around 7.5-fold fewer than bladder cancers ($P = 3.42 \times 10^{-4}$), which had more NUMTs than the other tumour types.

The average number of tumour-specific NUMTs did not correlate with age, implying that they arose during carcinogenesis, and not in somatic cells throughout life before cancer formation.

Tumour-specific NUMTs were more common on chromosome 19 ($P = 9.08 \times 10^{-6}$) and less common on chromosome 6 ($P = 1.53 \times 10^{-3}$) and were more likely to involve repetitive elements ($P = 4.24 \times 10^{-16}$), particularly satellite repeats ($P = 0.023$) and microsatellites repeats ($P = 0.007$) than the germline NUMTs. Finally, a greater proportion of tumour-specific NUMTs were found within 500 bp, 2,000 bp and 5,000 bp of the TSS than for germline NUMTs. Blunt-end repair in 27.6% of the NUMT breakpoints and short-nucleotide insertions in 20.5% of the NUMT breakpoints suggest the involvement of NHEJ.



Article

Nuclear-embedded mitochondrial DNA sequences in 66,083 human genomes

Consequences of NUMT insertion

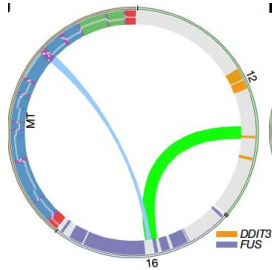
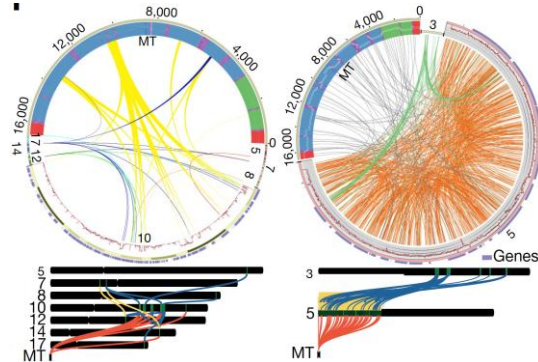
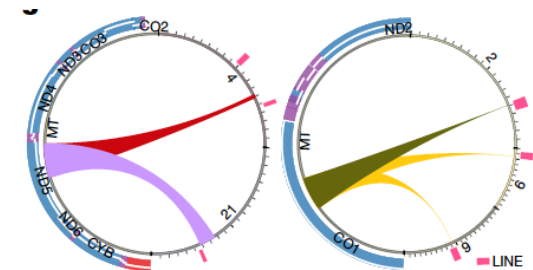
Nine hundred and forty six (58%) germline NUMTs were observed in gene regions, with the majority (85.8%, n = 812) being enriched in introns versus exons. No common or rare NUMTs ($F > 0.1\%$) were found in the coding DNA sequences (CDS) ($P = 0.039$ permutation test), and none were predicted to cause rare disease.

Two hundred and twenty tumour-specific NUMTs were found in gene regions, including 13 in CDS, 3 affecting stop codons, 4 affecting start codons, 16 in 3' or 5' UTRs. Eight tumours harboured tumour-specific NUMTs inserted into genes on the COSMIC Cancer Gene Census list (two in FHIT, which is a fragile genomic site, and one each in CTNNA2, DDIT3, WIF1, BCL11B, KDM5A and AKT2. One tumour had a NUMT insertion in an intron of FANCI, which is involved in DNA repair.

NUMTs are mobile after the initial insertion event.

Multiple fragments of the same mtDNA can be found in multiple chromosomes (mitochromothripsis).

Complex rearrangements with NUMT insertion at the site of chromosomal translocations were also seen in three out of eight tumour samples.



One myxoid liposarcoma tumour carried a FUS–DDIT3 chimeric fusion oncoprotein caused by a complex rearrangement involving a NUMT insertion.